

CHAPTER TWO

BASIC CONCEPTS OF LOGIC

2.1 Argument, Premises and Conclusions

Meaning: Logic is a science that evaluates arguments. We encounter arguments in our day-to-day experience. We hear them on TVs, read them in books, and formulate them when communicating with others. **The aim of Logic:** is to develop a system of methods/principles to evaluate arguments and to construct arguments of our own. Thus, Logic is a systematic study of methods for evaluating whether the premises of an argument adequately support (or provide good evidence for) the conclusion.

1.1. Conceptual Framework

I-A Statement: is a sentence/group of sentences that is either true or false, but not both (both at the same time)

II- An Argument: is a group of statements with one or more premises which are in turn claimed to support the conclusion. Two groups of arguments can be identified as:

1. Those in which the premises really support the conclusion- are called *Good arguments*, and
2. Those in which the premises do not support even though they are claimed to- are called *Bad arguments*.

So, Logic as a science evaluates arguments and allows us to distinguish good arguments from bad ones.

Statements: Consider these examples:

- Ex-1- Italy prevailed at Adwa in 1896. FS
- 2- Titanic was sunk by German sub marines in 1912. FS
- 3- Carrot is a good source of vitamin A. TS
- 4- Bishoftu is a city in Oromia. TS

Truth Values: examples 1 and 2 are false while 3 and 4 are true. So, truth and falsity are the two possible truth values of a statement. Unlike statements, many sentences can not be said to be true or false. These are non-statement expressions.

Non-Statements Expressions

1. Questions

Ex- What is your name?

2. Proposals

Ex- Let's go to the party today.

3. Suggestions

Ex- You would better go by bus.

4. Commands

Ex- Stand up!

Turn to the left at the next corner.

5. Exclamations

Ex- Bang up!

Right on!

The statements that make up an argument are divided into one or more premises and one and only one conclusion.

III- Premises: are statements that set forth reasons/evidence on the basis of which the conclusion is affirmed.

IV- Conclusion: is a statement that the evidence is claimed to support/imply. In other words, it is a statement that is claimed to follow from the premises.

N.B. The premise/s of an argument are offered as support or evidence for the conclusion, but that support may be *adequate or inadequate*. In a well-constructed argument, the premises give good support for believing that the conclusion is true. Consider these examples:

Ex-1- All crimes are violations of the law.TP

Theft is a crime.TP

Therefore, theft is a violation of the law.TC

Ex-2- All uncles are male. TP

Tola is an uncle. TP

So, Tola is a male. TC

In these examples, the premises really support the conclusion with good reasons. And the arguments are good ones.

But consider these examples:

Ex-3- Some crimes are misdemeanors. TP

Murder is a crime. TP

Therefore, murder is a misdemeanor. FC

In this argument, the premises do not support the conclusion, even though they are claimed to, and then the argument is bad.

Ex-4- Some uncles are skinny. TP

Gemechu is an uncle. TP

So, Gemechu is skinny. T/FC

Though the premises seem to support the conclusion, their evidence is very weak and the argument is a bad one.

Generally, any argument can be used:

- 1- To discover truth or
- 2- To persuade our readers/listeners.

Persuasion and truth seeking are often compatible goals. But one of these goals can interfere with the other.

2.2 How to differentiate a conclusion from its premise/s

Arguments usually contain indicator words that provide clues in identifying the premise/s and the conclusion.

1.1.1. Premise Indicators

Words like: Since, Because, As indicated by, May be inferred from, Owing to, In as much as, In that, For the reason that, Given that, Seeing that, As, For,...etc.

1.1.2. Conclusion Indicators

Words like: Therefore, Hence, So, Wherefore, Accordingly, Whence, It follows that, It must be that, Thus, As a result, We may infer, Consequently ...etc. The position of the

premise/s or the conclusion may change. Consider these:

Ex-1- Since the use of recreational drugs can jeopardize the development of the fetus; expectant mothers should never use these drugs.

Ex-2- The prevention of polio deserves increased expenditure in the years ahead. Not only polio affects unborn babies, but the virus still impinges up on the health status of already born babies.

In Ex-1 the premise comes first followed by the conclusion.

In Ex-2 the conclusion comes first followed by the premises.

2.3 Recognizing Arguments

Not all passages contain arguments. In general, a passage may contain an argument if it purports to prove some thing; if it does not do so, it does not contain an argument. Two conditions must be fulfilled for a passage to purport to prove some thing:

- 1- At least, one of the statements must claim to present evidence/reason and
- 2- There must be a claim that the alleged evidence/reason supports or implies some thing- that is, a claim that some thing follows from the alleged evidence.

It is not necessary that the premises present actual evidence or true reasons nor that the premises actually support the conclusion, but at least the premises must claim to present evidence or reason and there must be a claim that the evidence or reason supports or implies some thing.

The 1st condition- expresses a Factual Claim, that is, the truth or falsity of the premises.

The 2nd condition- an Inferential Claim which is the claim that the passage expresses a reasoning process or some thing follows from some thing/some thing supports some thing.

For factual claim, we ask "Are the premises all true?"

For inferential claim, we ask "Do the premises support the conclusion?"

Non-argument Expressions

Passages that lack an inferential claim are also called *Unsupported Assertions*. They are thus non-argument expressions. These include:

1. Warnings

Ex-1- Watch out that you do not slip on the mud.

2. Piece of Advice

Ex-1- I suggest you take Logic during your first year.

These two could serve as the conclusion of an argument, but in their present form/context there is no claim that they are supported or implied by reasons in evidence. Thus, there is no argument.

3. Statements of Belief/Opinion

These are expressions of what someone happens to believe in/think in a certain time.

Ex-1- I think a nation like ours, with its high moral traditions has a further responsibility to know how we become involved into this conflict, and to learn the lessons it has to teach us for the future.

4. Report

It consists of statements that convey information about some situation/event.

Ex-1- A bomb exploded near national museum, injuring 25 people and causing millions of birrs damage to the near by buildings, police said. A police statement said the 9000-birr bomb was packed into a basket hidden in the back of a stolen car.

Here no inferences are drawn and the passage merely contains a series of informational statements.

5. Illustration

It consists of statements about a certain subject combined with a reference to one or more specific instance/s intended to exemplify that statement. It is often confused with arguments because many illustrations contain indicator words such as "thus."

Ex.1. Mammals are animals that nourish their young with milk. For example, cats, dogs, goats, monkeys and humans are mammals.

Ex.2. Chemical elements, as well as, compounds, can be represented by molecular formulas. Thus, oxygen is represented by O₂, sodium chloride by NaCl and sulfuric acid by H₂SO₄.

Ex.3. Whole numbers can be represented as fractions. Thus, 2 can be represented as 8/4 and 5 can be represented as 15/3.

N.B. Sometimes examples are given not merely to explain but to support/provide

support for a thesis, in which case the passage becomes an argument and not an illustration.

Consider this:

Ex.1. You said that no mammal can fly, but that is untrue. At least one mammal has wings and can fly. For example, bats are mammals.

6. Conditional Statements

This is an “if..., then...” statement, taken by itself can never be an argument. Every conditional statement is made up of two statements. The component statement that immediately follows the “if” is called the Antecedent, and the one following the “then” is called the Consequent.

If..... then..... or

.....if.....

Ex.1. If Galgalo works hard, then he will get a promotion. This statement merely asserts that if Galgalo works hard, then he will get a promotion. So,

- a) It doesn't assert that he works hard and
- b) It doesn't assert that he gets a promotion.

By contrast consider this:

Ex.2. If Galgalo works hard, then he will get a promotion.

Galgalo has worked hard.

Therefore, Galgalo will get a promotion.

Here the conclusion is asserted on the basis of the premises. So the statement is an argument. While no conditional statement is an argument by itself, it can serve as either the premise or the conclusion (or both) of an argument.

Ex.3. If cigarette companies publish warning labels, then smokers assume the risk of smoking.

Cigarette companies do publish warning labels. Therefore, smokers assume the risk of smoking.

The relationship between a conditional statement and an argument can be summarized as:

- a) A single conditional statement can't be an argument,
- b) A conditional statement can serve as either the premise or the conclusion (or both) of an argument and
- c) The inferential content of a conditional statement may be re-expressed to form an argument.

Conditions in Conditional Statements

Conditional statements express the relationship between necessary and sufficient conditions.

- 'A' is said to be a **sufficient condition** for 'B' whenever the occurrence of 'A' is all that is needed for the occurrence of 'B'.

Ex.1. If I am stabbed by a dagger, then a scar will appear on my skin. (Sufficient condition for a scar can appear if I am stabbed by a bayonet, if I am burnt with a fire or if I am shot by a gun)

- On the other hand, 'B' is a **necessary condition** for 'A' whenever 'A' can't occur without the occurrence of 'B'.

Ex.1. If "X" is an uncle, then "X" is a male. (Necessary condition for one to be an uncle is one must be a male)

Ex.2. If water needs to be changed into an ice, then the temperature should be reduced below zero degree Celsius.

7. Explanations

These are a statement/group of statements intended to shed light on some phenomenon that is accepted as a matter of fact.

Every explanation consists of two distinct components: the Explanandum and the Explanans.

Explanandum: the statement that describes the event/phenomenon to be explained and

Explanans: the statement that purports to do the explaining.

Consider this example:

Ex.1. The sky appears blue from the earth's surface because light rays are scattered by particles from the atmosphere.

The intention of this example is to explain why the sky appears blue and not to prove

that it appears blue. So, it is not an argument.

Argument	Explanation
Premise/s -Accepted Fact/s	Explanans
Claimed to prove	Claimed to explain
Conclusion	Explanandum -Accepted Fact/s

But many explanations can be re-expressed to form an argument.

Ex.2. Light rays from the sun are scattered by particles in the atmosphere. Therefore, the sky appears blue from the earth's surface.

One last example: **Ex.3.** The dinosaurs are extinct because a large comet struck the earth some 6.5 million years ago, lofting a cloud of dust into the sky and blocking sunlight, thereby suppressing photosynthesis and... drastically lowering world temperatures.

(Explanation- so accepted fact which is scientific)

But: **Ex.4.** "Three scientific evidences have been offered for the extinction of dinosaurs. First, a global lowering in temperature caused the testes of male dinosaurs to stop functioning. Second, certain flowering plants (namely, angiosperms) evolved after the dinosaurs evolved; these plants were toxic for dinosaurs, which ate them and died. Thirdly,..." (Argument-Premises prove the Conclusion)

2.4 Types of Arguments

Arguments can be divided into two groups: *Deductive* and *Inductive* Arguments.

A Deductive Argument. is the one in which the premise/s are claimed to support its conclusion in such a way that if they are assumed true, it is impossible for the conclusion to be false. In such arguments the conclusion is claimed to follow necessarily from the premise.

Ex.1. All fruits are beverages. TP
Oranges are fruits TP
So, oranges are beverages.TC

Ex.2. All mammals are vertebrates.TP
All humans are mammals.TP
Therefore, all humans are vertebrates.TC

Ex.3. All saleswomen in this super market are hospitable.TP

Marartu is a saleswoman in this super market.TP

Therefore, Marartu is hospitable.TC

In these three arguments the conclusion follows from the premises with strict necessity.
So, the arguments are deductive.

An Inductive Argument: is an argument in which the premise are claimed to support to conclusion in such a way that if they assumed true, then based on that assumption it is improbable that the conclusion is false(that is, it is probable that the conclusion is true). In these arguments the conclusion is claimed to follow only probably from the premises.
Consider these examples:

Ex.1. The vast majority of the saleswomen in this super market are hospitable.TP

Obse is a saleswoman in this super market.TP

Therefore, probably, Obse is hospitable.TC

Ex.2. 80 oranges selected at random from a basket containing 100 oranges were found to be ripe.TP

Probably, all the 100 oranges are ripe.TC

I/- These arguments are inductive because the occurrence of the conclusion is based on probable/improbable conditions.

Sub categories of Deductive Argument

A) Arguments based on mathematics

These are argument in which their conclusion depends on some purely arithmetic/geometric computation/measurements.

Ex.1 a shopper might place two apples and three oranges into a paper bag and conclude that the bag contains five pieces of fruit.

B. An argument from definition

An argument in which the conclusion is claimed to depend merely up on the definition of some word or phrase used in the premises or conclusion. Some might argue that Michael is mendacious, it follow that he tells lies, or that because a certain paragraph is prolix, it follows that it is excessively wordy.

C) Categorical Syllogism

This a syllogism in which each statement begins with one of the words “all”, “no” or “some”.

Ex.1. No cats are dogs.TP

No dogs are mammals that can fly.TP

Therefore, no cats are mammals that can fly.TC

D) Hypothetical Syllogism

This is a syllogism having a conditional statement for one or both of its premises.

Ex.1. If you trespass campus compound illegally, then the campus police will catch you.TP

If the campus police catch you, then you will face disciplinary penalty.TP

Therefore, if you trespass the campus compound illegally, then you will face disciplinary penalty. TC

Ex.2. If a stone scratches glass, then it is harder than the glass.TP

Stone scratches glass.TP

So, stone is harder than glass.TC

E) Disjunctive Syllogism

This is a syllogism having a disjunctive statement of an “either...or...” statement for one of its premises.

Ex.1. Either breach of contract is a crime or it is not punishable by law.TP

Breach of contract is not a crime.TP

So, breach of contract is not punishable by law.TC

//- Sub categories of Inductive Argument

A) Arguments from Analogy

These are the ones that depend on the existence of analogy between two things.

Ex.1. Abebe’s 2006 model Chevoloret has luxurious seats, an excellent gas mileage and a computer set. TP

Matiyas’s 2006 model Chevoloret has luxurious seats, and an excellent gas mileage.TP
Probably, Mati yas’s Chevoloret has a computer set. T/FC

B) Inductive Generalization

This is an argument that proceeds from the knowledge of a selected sample to some claim about the whole group.

Ex.1. 10 apples selected at random from a basket containing 100 apples were found to be ripe. TP probably, all the rest 90 apples are ripe. FC

C) Arguments based on Signs

If signs like traffic symbols are placed or misplaced from the right position, then the conclusion rests on chances of probability/improbability.

D) A Causal Inference

- 1) It underlies arguments that proceed from the knowledge of a cause to the knowledge of an effect or
- 2) It underlies from the knowledge of an effect to knowledge of a cause.

Ex.1. A bottle of wine was accidentally put in the freezer. Therefore, the bottle of wine had been frozen. (Cause to Effect)

Ex.2. I have tasted a piece of chicken and found it dry and crunchy. Therefore, the chicken had been overcooked. (Effect to Cause)

2.5 Evaluating Arguments

Deductive Argument

Every argument makes two basic claims: a claim that the evidence exists which is called factual claim and a claim that the alleged evidence supports something (something follows from the alleged evidence). The most crucial of the two is an inferential claim because if the premises fail to support the conclusion (that is, if the reasoning is bad), an argument is worthless. Thus, we will always test the inferential claim first, and only if the premises support the conclusion will we test the factual claim.

There are two types of deductive argument: Valid and Invalid deductive arguments.

I. Valid Deductive Argument

This is an argument such that if the premises are assumed true, it is impossible for the conclusion to be false. In such a case, the conclusion follows with strict necessity from the premises.

II Invalid Deductive Argument

This is the one such that if the premises are assumed true, it is possible for the conclusion to be false. Here the conclusion does not follow with strict necessity from the premises.

Mind you! **First**, there is no middle ground between valid and invalid arguments. As such there are no “almost valid” or “almost invalid” arguments. **Second**, there is only indirect relationship between validity and truth. For an argument to valid, it is not necessary that either the premises/conclusion or both be true, but merely that if the premises are assumed true, it is impossible for the conclusion to be false.

SCENARIOS	VALID	INVALID
True premise True Conclusion	All wines are beverages. TP Gouder is a wine. TP Gouder is a beverage. TC SOUND	All wines are beverages.TP Gouder is a beverage. TP So, gouder is a wine. TC UNSOUND
True Premise False Conclusion	NON-EXIST <i>Because in deductive argument, to be valid if premise assumed true the conclusion necessarily needs to be true.</i>	All wines are beverages.TP Orange juice is a beverage.TP So, orange juice is a wine. FC UNSOUND
False premise True Conclusion	All wines are soft drinks. FP Coca Cola is a wine. FP So,CocaCola is a soft drink.TC UNSOUND	All wines are whiskeys. FP Gouder is a whiskey. FP So, gouder is a wine. TC UNSOUND
False premise False conclusion	All wines are whiskeys. FP Coca Cola is a wine. FP So, Coca Cola is a whiskey. FC UNSOUND	All wines are whiskeys. FP Coca Cola is a whiskey. FP So, Coca Cola is a wine. FC UNSOUN

///. Soundness and Unsoundness of Arguments

A Sound Argument: is a deductive argument that is valid and has all true premises. Every sound argument by definition has true conclusion. It is meant a “good” deductive argument.

A Sound deductive Argument= Valid + All True Premises

Inductive Argument

Here, the premises are claimed to support the conclusion with certain degree of probability or improbability depending on the strength or weakness of the reasoning or presented evidence. There are two types: Strong and Weak inductive arguments.

I. Strong Inductive Argument

It is the one such that if the premises are assumed true, then it is probable that the conclusion is false.

Weak Inductive Argument

It is the one such that if the premises are assumed true, then it is not probable that the conclusion is false.

SCENARIOS	STRONG	WEAK
True premise True Conclusion	All previous American presidents were men. Therefore, probably the next American president will be a man. COGENT	A few American presidents were Federalists. Therefore, probably the next American president will be a man. UNCOGENT
True Premise False Conclusion	NONE EXIST	A few American presidents were Federalists. Therefore, probably the next American president will be a Federalist. UNCOGENT
False premise True Conclusion	All previous American presidents were television debaters. Therefore, probably the next American president will be a television debater. UNCOGENT	A few American presidents were Libertarians. Therefore, probably the next American president will be a television debater. UNCOGENT
False premise False conclusion	All previous American presidents were women. Therefore, probably the next American president will be a woman. UNCOGENT	A few American presidents were Libertarians. Therefore, probably the next American president will be a Libertarian. UNCOGENT

II. Cogency and Uncogency of Inductive Arguments

A Cogent Inductive Argument: is the one that is strong and has all true premises. It is an inductive analogy of a sound deductive argument. It is meant to be “a good” inductive

argument. The conclusion of every cogent argument is probably true.

Cogent= Strong + All True Premises

There is a difference between sound and cogent arguments in regard to true premises requirement. In a sound argument it is only necessary that the premises be true. In a cogent argument, on the other hand, the premises must not only be true, they must also ignore some important piece of evidence that outweighs the given evidence.

Uncogent Argument falls into one of the following **three** categories:

- a) It is strong but has at least one (more) false premise/s.
- b) It is weak but has all true premises.
- c) It is weak but has at least one (more) false premise/s.